Telecom Customer Churn Prediction

Machine Learning Analysis Report

7,010	26.49%	0.64
Records	Churn Rate	Best Recall

Prepared by: Data Science Team

Dataset: IBM Telco Customer Churn | Source: Kaggle

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1. Executive Summary

The objective of this analysis was to build a predictive model to identify customers who are likely to churn so the company can take proactive retention actions. Exploratory data analysis (EDA) revealed a significant class imbalance, with far fewer churners than non-churners. Because of this imbalance, **recall** was used as the primary evaluation metric to ensure the models focus on correctly identifying customers who actually churn.

Several machine learning models — KNN, Random Forest, and Decision Tree — were developed and evaluated using hyperparameter tuning via GridSearchCV. The **Decision Tree** achieved the highest recall on the test set (0.64), indicating it is the most effective at identifying at-risk customers while generalizing well to unseen data.

> **Recommendation:** Deploy the Decision Tree model (test recall = 0.64) to power targeted retention campaigns. Focus early-lifecycle engagement (first 90 days) to move month-to-month customers into longer contract tiers and integrated service bundles.

2. Dataset Overview

The dataset is the *IBM Watson Telco Customer Churn* dataset sourced from Kaggle. After removing 22 duplicate records and 11 null values, the final working dataset contained **7,010 rows and 21 columns**.

7,010	21	26.49%	22
Clean Records	Features	Churn Rate	Duplicates Removed

2.1 Data Dictionary

Variable	Description
CustomerID	Unique customer identifier
Gender	Customer gender (Male / Female)
SeniorCitizen	Whether customer is a senior citizen (1 = Yes, 0 = No)
Partner	Whether customer has a partner (Yes / No)
Dependents	Whether customer has dependents (Yes / No)

Tenure	Number of months with the company
PhoneService	Whether customer has phone service
MultipleLines	Whether customer has multiple lines
InternetService	Internet service provider (DSL, Fiber optic, No)
OnlineSecurity	Whether customer has online security add-on
OnlineBackup	Whether customer has online backup add-on
DeviceProtection	Whether customer has device protection add-on
TechSupport	Whether customer has tech support add-on
StreamingTV	Whether customer streams TV
StreamingMovies	Whether customer streams movies
Contract	Contract term (Month-to-month, One year, Two year)
PaperlessBilling	Whether customer uses paperless billing
PaymentMethod	Payment method used
MonthlyCharges	Monthly charge amount ($)
TotalCharges	Total charges to date ($)
Churn	Whether customer churned (Yes / No) — Target Variable

3. Data Preprocessing

- **Removed CustomerID** — no analytical value.
- **Dropped 22 duplicate rows** and **11 null TotalCharges values** after type conversion.
- **Outlier removal** using the IQR method on Tenure, MonthlyCharges, and TotalCharges.
- **Binary encoding** of Yes/No features; 'No internet service' and 'No phone service' collapsed to 'No'.
- **One-hot encoding** of PaymentMethod, Contract, and InternetService (drop-first).
- **StandardScaler** normalization applied to Tenure, MonthlyCharges, and TotalCharges.
- **80/20 train-test split** with random_state = 42.

Descriptive Statistics (numeric features)

Statistic	SeniorCitizen	Tenure	MonthlyCharges	TotalCharges

Count	7,010	7,010	7,010	7,010
Mean	0.16	32.52	64.89	2,290.35
Std	0.37	24.52	30.06	2,266.82
Min	0.00	1.00	18.25	18.80
25%	0.00	9.00	35.75	408.31
50%	0.00	29.00	70.40	1,403.88
75%	0.00	56.00	89.90	3,807.84
Max	1.00	72.00	118.75	8,684.80

4. Exploratory Data Analysis

4.1 Customer Demographics

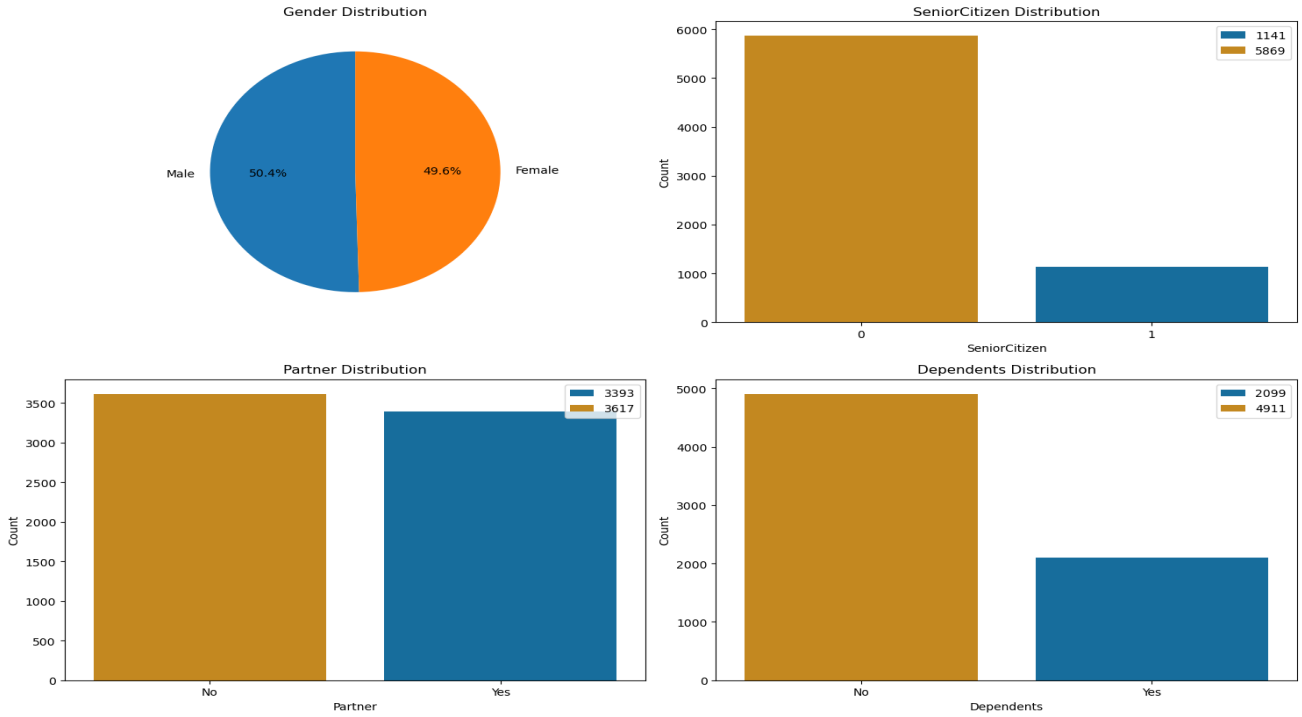

Figure 1 — Customer demographics distribution (gender, senior citizen, partner, dependents)

The customer base is almost evenly split by gender (~50/50). The majority are not senior citizens, suggesting the service primarily attracts working-age adults. Slightly more customers have no partner, and most customers have no dependents, indicating the base is largely composed of independent adults.

4.2 Billing & Charges

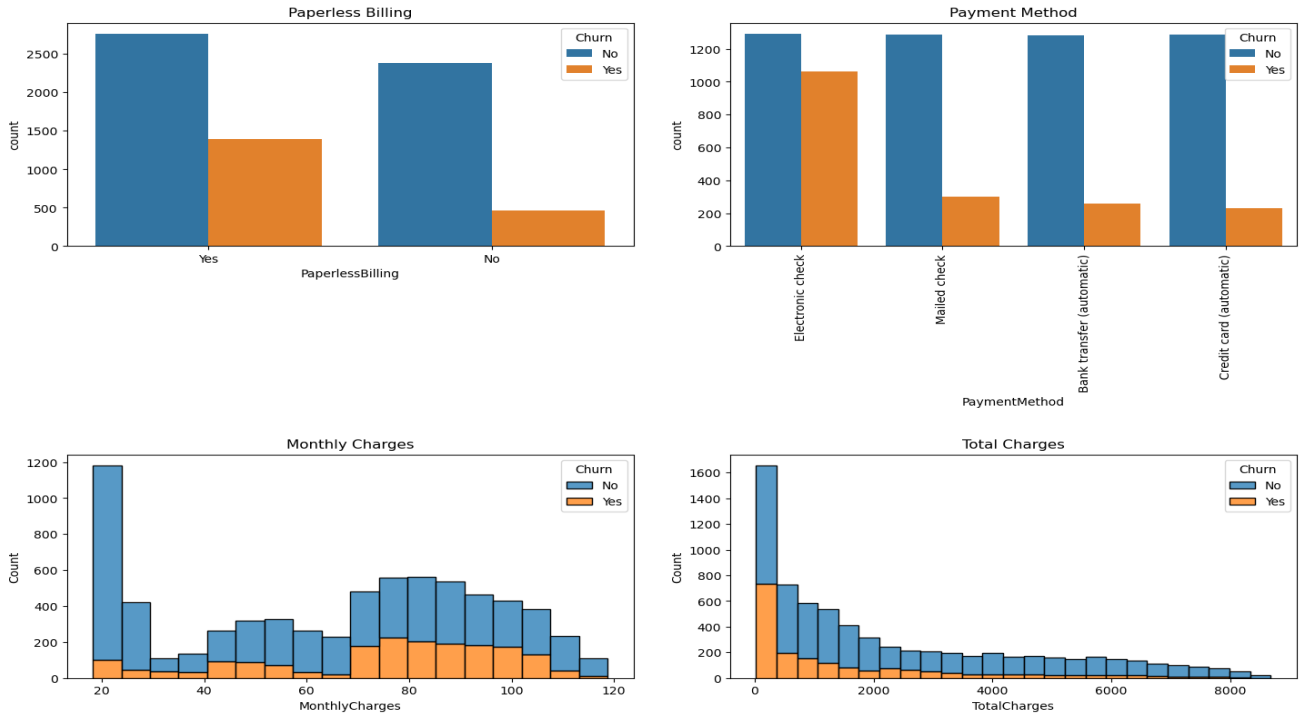

Figure 2 — Billing and charges distribution

A large spike at low Monthly Charges indicates a sizable segment on basic plans. Many of these customers pay via Electronic Check — a low-friction method associated with higher churn. The concentration of Total Charges under $500 aligns with the early-tenure churn spike, confirming that short-tenure, low-cost users are a revolving-door segment.

4.3 Key Churn Drivers (Correlation with Target)

Pearson correlations reveal the features most associated with churn:

Feature (Negative Corr. w/ Churn)	r
tenure	-0.35
Contract_Two year	-0.30
InternetService_No	-0.23
TotalCharges	-0.20
OnlineSecurity	-0.17
Feature (Positive Corr. w/ Churn)	**r**
InternetService_Fiber optic	+0.31

PaymentMethod_Electronic check	+0.30
MonthlyCharges	+0.19
PaperlessBilling	+0.19
SeniorCitizen	+0.15

> **Key insight:** Longer tenure and two-year contracts are the strongest protective factors against churn. Fiber optic internet and electronic check payment are the strongest risk indicators.

5. Candidate Modelling

Three classifiers were trained and tuned using GridSearchCV with 3-fold cross-validation, optimizing for **recall** given the class imbalance (73.51% non-churn vs. 26.49% churn).

5.1 Hyperparameter Tuning Results

Model	Best Hyperparameters
Decision Tree	criterion=entropy, max_depth=6, min_samples_leaf=6, min_samples_split=2, random_state=0
Random Forest	criterion=entropy, max_depth=10, min_samples_leaf=2, min_samples_split=6, n_estimators=300, random_state=0
KNN	algorithm=auto, n_neighbors=10, weights=distance

6. Model Evaluation

6.1 Recall Comparison

Model	Train Recall	Test Recall	Overfit?	Recommended
Decision Tree	0.65	0.64	No	YES
Random Forest	0.63	0.53	Moderate	—
KNN	0.99	0.54	YES	—

The KNN model severely overfits (train recall 0.99 vs. test 0.54). The Random Forest shows moderate overfitting. The Decision Tree demonstrates the most consistent performance across train (0.65) and test

(0.64) sets.

6.2 Confusion Matrices

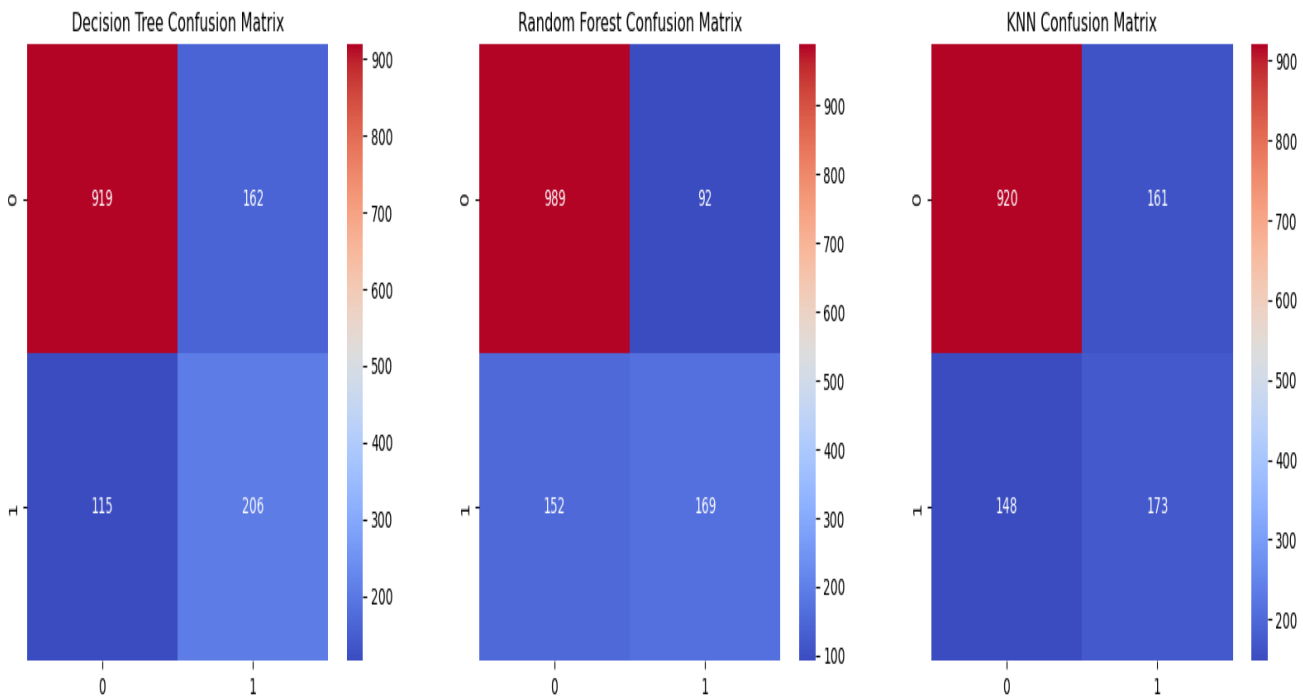

Figure 3 — Confusion matrices for Decision Tree, Random Forest, and KNN

The Decision Tree identifies **206 True Positives** (actual churners correctly flagged) with only **115 False Negatives** (missed churners). Minimizing False Negatives is critical in telecom churn: every undetected churner represents a lost retention opportunity. The Random Forest produces more True Negatives but misses significantly more churners.

6.3 Feature Importance

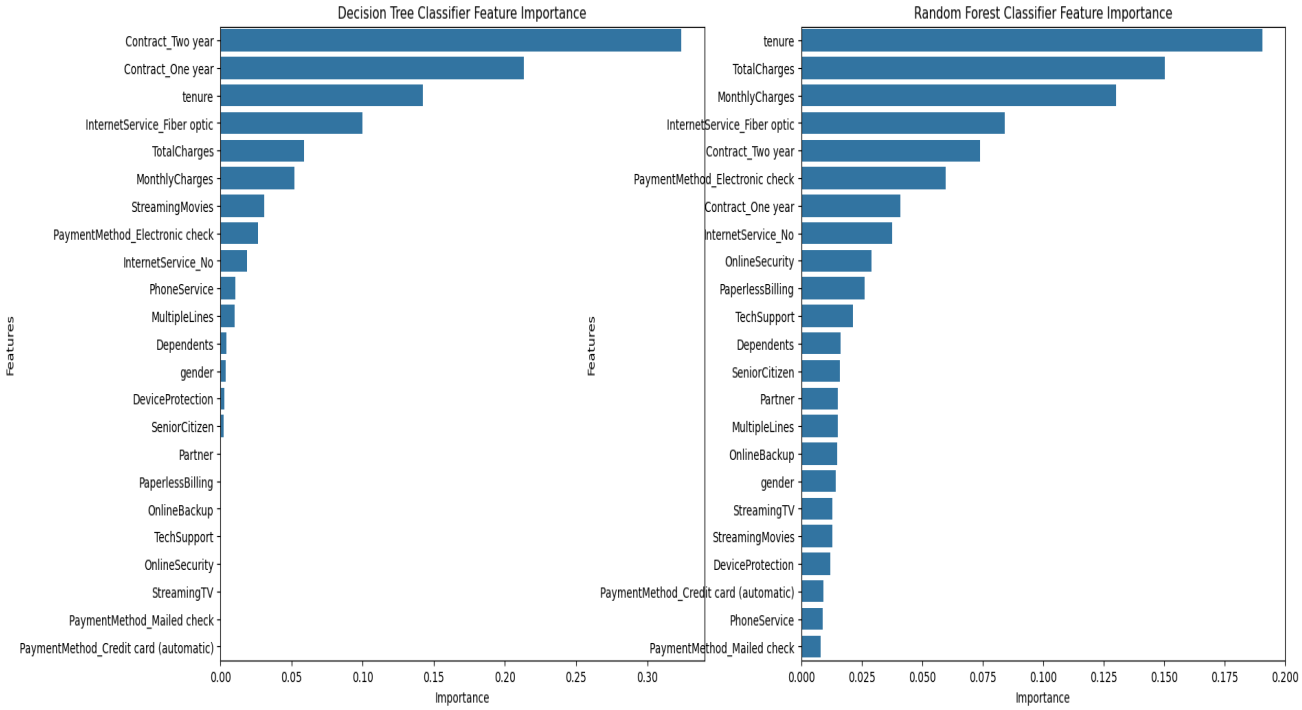

Figure 4 — Feature importance for Decision Tree (left) and Random Forest (right)

Both models agree on the top predictors: **contract type, tenure, and pricing variables** (monthly and total charges). The Decision Tree additionally highlights one-year and two-year contracts as the strongest retention signals. These findings align with the EDA correlation analysis.

7. Conclusions & Recommendations

The Decision Tree model is recommended as the production model for identifying at-risk customers. It achieves a test recall of 0.64 — the highest among all evaluated models — while generalizing well with minimal overfitting.

Strategic Recommendations

- **Target early-lifecycle customers (0–90 days).** The early-tenure churn spike is the single biggest retention opportunity. Onboarding campaigns and service-bundle incentives during the first 90 days can anchor new customers.
- **Incentivize contract upgrades.** Month-to-month customers are the highest-risk segment. Promotions offering discounts for switching to one- or two-year contracts should be prioritized.
- **Promote value-added services.** Customers without Online Security, Tech Support, or Device Protection churn at higher rates. Cross-selling these services raises switching costs and deepens platform engagement.

- **Monitor fiber optic + electronic check customers.** This combination shows the highest churn risk. Targeted outreach and loyalty rewards for this segment can reduce attrition.
- **Reduce monthly charge sensitivity.** High monthly charges correlate positively with churn. Flexible pricing tiers or loyalty discounts for long-tenure customers may help retain price-sensitive segments.

> **Model Performance Summary:** Decision Tree — Train Recall 0.65 | Test Recall 0.64 | True Positives 206 | False Negatives 115. Recommended for deployment in the customer retention pipeline.